

Minimal Replacement Blockers in SG-to-GFL Grid Transitions: Exact Collective Closure, Compatibility Geometry, and Critical Green-Residue Repair

Jorge L. Mayorga Taborda
Universidad de los Andes, Bogota, Colombia

Abstract—Replacing synchronous generators (SGs) by grid-following converters (GFLs) is often summarized by penetration, grid-strength, or fixed-composition stability metrics. This manuscript develops a different planning object: an inclusion-minimal SG-to-GFL replacement portfolio whose transverse dynamics are unstable although every proper replacement subset is stable. Starting from the full phasor-domain differential-algebraic equations (DAEs), we derive an exact transverse quotient, a dynamic multiport representation, and a determinant factorization that separates local replacement factors from network-mediated collective closure. At a simple collective boundary, a critical eigenvalue of the interaction operator reaches -1 , while every blocker member sees the remainder through a contextual return with eigenvalue $+1$. In the frozen IEEE 39-bus benchmark, all 15 proper subsets of $\{30, 33, 35, 37\}$ are stable while the complete portfolio has $\alpha_{\perp} = +0.1270 \text{ s}^{-1}$ at 0.6223 Hz ; physical local factors remain regular while the complete collective singular value falls to 1.26×10^{-8} . We distinguish the complete-spectrum blocker hypergraph $\mathcal{H}_{\perp}$ from a mode-scoped diagnostic hypergraph $\mathcal{H}_{\Omega}$, because a hard frequency window can change membership without an imaginary-axis crossing. Controller policy reorganizes the complete-spectrum blocker geometry, showing that compatibility is not a scalar penetration number. Finally, we replace a singular “Green function at the pole” by the critical resolvent residue $\mathcal{G}_H^*$, derive the total pole sensitivity including power-flow re-equilibration, and formulate blocker-conditioned line, bus, zone, and hypergraph-aware repair scores. These repair equations are derived here, but their blind 46-branch ranking experiment remains a preregistered validation target rather than a claimed result. The manuscript is a detailed theory and presentation companion, not a universal stability or EMT certificate.

Index Terms—Inverter-based resources, grid-following converters, small-signal stability, nonlinear eigenvalue problem, Schur complement, Green function, dynamic graph, grid strength, minimal failure set, topology repair, IEEE 39-bus system.

I. MOTIVATION AND FOUR-RESULT NARRATIVE

THE central engineering question is not only how much inverter-based generation a grid can host, but which *finite combinations* of technology substitutions first become dynamically incompatible. Existing small-signal, impedance, grid-strength, critical-cluster, and stability-manifold methods answer important neighboring questions [1], [3], [4], [6]–[8]. The present program asks instead:

Working technical companion manuscript for the IEEE IAS Annual Meeting 2026 poster development. It consolidates proved theory, frozen IEEE-39 evidence, and one explicitly labeled proposed extension for dynamic topology repair.

Which inclusion-minimal SG-to-GFL replacement sets fail, what exact dynamic feedback creates their boundary, how does controller policy reorganize the failing sets, and which control or topology actions move the boundary most effectively?

The poster-oriented theory is organized around four results:

- 1) **Minimal blocker:** a complete replacement set can be unstable although every proper subset is stable.
- 2) **Exact collective closure:** local device factors can remain regular while a network-mediated interaction factor alone becomes singular.
- 3) **Compatibility geometry:** the minimal incompatible sets form a policy-dependent hypergraph, not a single penetration threshold.
- 4) **Diagnose-to-repair:** the same dynamic operator yields exact boundary derivatives and, after a critical Green-residue lift, line/bus/zone repair fields.

The novelty position is intentionally narrow. Minimal failure sets, Schur complements, generalized Nyquist/return ratios, eigenvalue sensitivities, and resolvent residues are classical or have direct prior use in power-system impedance participation and remedial-action studies [5], [9]–[11]. The candidate contribution is their synthesis around a counterfactual SG→GFL replacement hypergraph: identify a minimal failing portfolio, prove its collective-return anatomy, track how controller policy reorganizes that discrete object, and design repairs that protect the hypergraph rather than only one fixed composition.

Table I distinguishes proved/validated statements from the design extension that still requires its dedicated branch-ranking experiment.

II. PHYSICAL POWER-SYSTEM MODEL

A. AC network and index-one DAE

Let the network have N buses and complex voltages

$$V_i = V_{Ri} + jV_{Hi} = |V_i|e^{j\theta_i}. \quad (1)$$

With $Y_{\text{bus}} = G + jB$, Kirchhoff current balance may be written in rectangular coordinates as

$$g(\mathbf{x}, \mathbf{z}; \delta, \theta) = \mathcal{Y}_{\text{bus}} \mathbf{z} - \mathbf{i}_{\text{dev}}(\mathbf{x}, \mathbf{z}; \delta, \theta) + \mathbf{i}_L(\mathbf{z}) = \mathbf{0}, \quad (2)$$

TABLE I

CLAIM LEDGER: MATHEMATICAL STATUS, NOVELTY POSITIONING, AND EVIDENCE. “NEW” BELOW REFERS TO THE PRESENT RESEARCH PROGRAM, NOT TO THE UNDERLYING CLASSICAL ALGEBRAIC IDENTITIES.

Result	Mathematical status	Novelty positioning	Evidence/status
Transverse quotient and symmetry removal	PROVED	Classical symmetry reduction specialized to the frozen model	Exact identity; numerical audit
Minimal replacement blocker $\mathcal{H}$	Exact definition / elementary structure	New planning object/application; antichain mathematics is classical	Frozen V4/V9 and policy maps
Network-closure factorization	PROVED	Classical Sylvester/Schur algebra; new SG→GFL portfolio interpretation	Identity residual $\leq 1.4 \times 10^{-13}$
Contextual return at a blocker	PROVED	Classical Schur return identity; new blocker-conditioned interpretation	TX4 sign/residual audit
Policy-dependent compatibility geometry	Structural theorem + empirical map	Candidate substantive contribution	30/36/16 distinct hypergraphs
Connected cumulants	PROVED secondary layer	Classical partition-cumulant machinery adapted to finite portfolios	Flagship found composite ($\nu_H = 0.069$)
Critical Green residue / total repair derivative	PROVED from NEP perturbation theory	Residue-sensitivity link has prior art; novelty target is blocker-conditioned, hypergraph-aware repair	DERIVED; blind branch-ranking test pending
Portfolio-level EMT	—	—	NOT ESTABLISHED

where

$$\mathcal{Y}_{\text{bus}} = \begin{bmatrix} G & -B \\ B & G \end{bmatrix}, \quad \mathbf{z} = [V_R^\top \ V_I^\top]^\top. \quad (3)$$

Constant-power loads satisfy

$$I_{L,i} = \frac{P_{L,i} - jQ_{L,i}}{V_i^*}. \quad (4)$$

Equivalently, for $V_i = a_i + jb_i$,

$$I_{L,R,i} = \frac{P_{L,i}a_i + Q_{L,i}b_i}{a_i^2 + b_i^2}, \quad (5)$$

$$I_{L,I,i} = \frac{P_{L,i}b_i - Q_{L,i}a_i}{a_i^2 + b_i^2}. \quad (6)$$

The full phasor model is the index-one DAE

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{z}; \delta, \theta), \quad (7)$$

$$\mathbf{0} = \mathbf{g}(\mathbf{x}, \mathbf{z}; \delta, \theta), \quad (8)$$

where $\delta \in \{0, 1\}^m$ is the replacement indicator and θ collects controller-policy coordinates. At a regular equilibrium, $\mathbf{g}_z$ is nonsingular and the usual reduced Jacobian is

$$\mathbf{A} = \mathbf{f}_x - \mathbf{f}_z \mathbf{g}_z^{-1} \mathbf{g}_x. \quad (9)$$

B. Two-axis synchronous generator, AVR, and PSS

For each synchronous unit, let $v_d + jv_q = V_i e^{-j\delta_i}$ and let i_d, i_q be obtained from the transient-reactance stator equations. The frozen model uses

$$\dot{\delta}_i = \omega_B(\omega_i - 1), \quad (10)$$

$$M_i \dot{\omega}_i = P_{m,i} - P_{e,i} - D_i(\omega_i - 1), \quad (11)$$

$$T'_{d0,i} \dot{e}'_{q,i} = e_{fd,i} - e'_{q,i} - (x_{d,i} - x'_{d,i})i_{d,i}, \quad (12)$$

$$T'_{q0,i} \dot{e}'_{d,i} = -e'_{d,i} + (x_{q,i} - x'_{q,i})i_{q,i}. \quad (13)$$

The electrical power is

$$P_{e,i} = v_{d,i}i_{d,i} + v_{q,i}i_{q,i} + r_{a,i}(i_{d,i}^2 + i_{q,i}^2). \quad (14)$$

The first-order AVR and power-input stabilizer are

$$T_{E,i} \dot{e}_{fd,i} = K_{A,i}(v_{\text{ref},i} + v_{s,i} - |V_i|) - e_{fd,i}, \quad (15)$$

$$T_{6,i} \dot{x}_{5,i} = P_{e,i} - x_{5,i}, \quad (16)$$

$$T_{4,i} \dot{x}_{6,i} = \frac{T_{5,i}}{T_{6,i}}(P_{e,i} - x_{5,i}) - x_{6,i}, \quad (17)$$

$$v_{s,i} = K_{S,i}x_{6,i}. \quad (18)$$

The frozen benchmark has $D_i = 0$ and constant $P_{m,i}$; therefore it contains no primary frequency restoration. This is the reason transverse coordinates are required later.

C. Eleven-state grid-following converter

The GFL state vector is

$$\mathbf{x}_{\text{gfl}} = [\vartheta, x_{\text{pll}}, p_f, q_f, x_p, x_q, i_d, i_q, x_{id}, x_{iq}, x_v]^\top. \quad (19)$$

With $v_d + jv_q = V_i e^{-j\vartheta}$ and

$$P = v_d i_d + v_q i_q, \quad Q = v_q i_d - v_d i_q, \quad (20)$$

the PLL is

$$\dot{\vartheta} = k_{p,\text{pll}} v_q + x_{\text{pll}}, \quad (21)$$

$$\dot{x}_{\text{pll}} = k_{i,\text{pll}} v_q. \quad (22)$$

Filtered powers obey

$$\tau \dot{p}_f = P - p_f, \quad (23)$$

$$\tau \dot{q}_f = Q - q_f. \quad (24)$$

The outer PI loops are

$$\dot{i}_d^{\text{ref}} = k_p(p_{\text{ref}} - p_f) + x_p, \quad (25)$$

$$\dot{i}_q^{\text{ref}} = -k_q(q_{\text{cmd}} - q_f) - x_q, \quad (26)$$

$$\dot{x}_p = k_{ip}(p_{\text{ref}} - p_f), \quad (27)$$

$$\dot{x}_q = k_{iq}(q_{\text{cmd}} - q_f). \quad (28)$$

The current loops are

$$\dot{i}_d = \frac{\omega_B}{x_f} [k_i(i_d^{\text{ref}} - i_d) + x_{id} - r_f i_d], \quad (29)$$

$$\dot{i}_q = \frac{\omega_B}{x_f} [k_i(i_q^{\text{ref}} - i_q) + x_{iq} - r_f i_q], \quad (30)$$

$$\dot{x}_{id} = k_{ii}(i_d^{\text{ref}} - i_d), \quad (31)$$

$$\dot{x}_{iq} = k_{ii}(i_q^{\text{ref}} - i_q). \quad (32)$$

TABLE II
CORE FROZEN DEVICE PARAMETERS (PER-UNIT ON DEVICE RATING).

Block	Values
SG	two-axis, $D = 0$, constant P_m
AVR	$K_A = 10.1$, $T_E = 0.25$ s
PSS	$K_S = -2$, $T_5 = 1$, $T_6 = 4.2$, $T_4 = 0.75$ s
PLL	$k_{p,pll} = 53$, $k_{i,pll} = 1400$
Outer loops	$\tau = 0.03$ s, $k_p/k_{ip} = k_q/k_{iq} = 0.2/8$
Current loops	$k_i/k_{ii} = 0.25/6$, $x_f = 0.15$, $r_f = 0.01$
Q/V	$k_{p,v}/k_{i,v} = 2/20$, $w = 0.05$ rad/s

The leaky Q/V regulator is

$$e_v = v_{\text{ref}} - |V|, \quad (33)$$

$$q_{\text{cmd}} = k_{p,v} g e_v + x_v, \quad (34)$$

$$\dot{x}_v = k_{i,v} g e_v - w(x_v - q_{\text{ref}}). \quad (35)$$

The current injected into the network is

$$I_i^{\text{gf}} = \gamma_i (i_d + j i_q) e^{j\theta_i}. \quad (36)$$

The principal frozen values are listed in Table II.

D. Matched-dispatch replacement portfolio

Let the candidate set be $\mathcal{V} = \{1, \dots, m\}$. A portfolio $S \subseteq \mathcal{V}$ has indicator δ_S . Replacing unit i installs a GFL with the same scheduled P/Q and the same rating as the removed SG; the AC power-flow equilibrium is therefore held fixed across the discrete portfolio family. The frozen IEEE-39 core is

$$\mathcal{V}_4 = \{30, 33, 35, 37\}, \quad (37)$$

and the nine-unit census uses generator buses 30–38. The term *replacement transition* in this manuscript refers to a sequence of re-equilibrated stationary configurations, not to the fast switching transient of physically disconnecting an SG and energizing an inverter. The policy vector is

$$\theta = (g, k, t, h), \quad (38)$$

with g the GFL Q/V gain, k a common surviving-SG AVR gain scale, t an AVR time-constant scale, and h a heterogeneity exponent.

III. EXACT TRANSVERSE STABILITY

The frozen model has a structural rotational symmetry and, because $D = 0$ and no governor is present, an additional frequency-drift Jordan partner. Directly deleting “small” eigenvalues by magnitude is therefore not an exact stability definition.

Let R_x be one on every rotor/PLL angle coordinate and zero elsewhere. Let w represent a uniform frequency shift.

Theorem 1 (Exact transverse quotient): Assume device angles enter the DAE only through rotation-covariant combinations $V e^{-j(\cdot)}$. Then

$$A R_x = 0. \quad (39)$$

If, in addition, synchronous damping is zero, mechanical power is constant, and the network is frequency independent, then

$$A w = \omega_B R_x. \quad (40)$$

Hence $C = \text{span}\{R_x, w\}$ is A -invariant. For any orthonormal Z spanning $C^\perp$,

$$A_\perp = Z^T A Z \quad (41)$$

and

$$\det(sI - A) = s^2 \det(sI - A_\perp). \quad (42)$$

Proof: Rotation covariance gives, for an infinitesimal common phase ϕ ,

$$f(x + \phi R_x, e^{j\phi} z) = f(x, z), \quad (43)$$

$$g(x + \phi R_x, e^{j\phi} z) = 0. \quad (44)$$

Differentiate with respect to ϕ at the equilibrium. With $R_z = jV$,

$$f_x R_x + f_z R_z = 0, \quad (45)$$

$$g_x R_x + g_z R_z = 0. \quad (46)$$

Because g_z is nonsingular, $R_z = -g_z^{-1} g_x R_x$, and substitution into $A = f_x - f_z g_z^{-1} g_x$ yields $A R_x = 0$. Under the additional no-damping/no-governor assumptions, a uniform speed offset c together with the phase ramp $c \omega_B t R_x$ generates another exact solution family; differentiation in c gives $A w = \omega_B R_x$. In the orthogonal basis $[U \ Z]$ with U spanning C , invariance implies $Z^T A U = 0$, so the transformed matrix is block upper triangular with the 2×2 nilpotent Jordan block on C and $A_\perp$ on $C^\perp$. Equation (42) follows. ■

Definition 1 (Complete-spectrum transverse abscissa):

$$\alpha_\perp(S, \theta) = \max_{\lambda \in \text{spec}(A_\perp(S, \theta))} \text{Re } \lambda. \quad (47)$$

A portfolio is transversely stable when $\alpha_\perp < 0$.

Definition 2 (Mode-scoped diagnostic abscissa): For a reporting band Ω ,

$$\alpha_\Omega(S, \theta) = \max_{\lambda \in \text{spec}(A_\perp): |\text{Im } \lambda|/(2\pi) \in \Omega} \text{Re } \lambda. \quad (48)$$

For the poster mechanism, $\Omega = [0.3, 1.5]$ Hz.

The distinction is essential. $\alpha_\perp$ is the stability label used for complete-spectrum blocker geometry. α_Ω is a mode-scoped diagnostic used to isolate the validated electromechanical mechanism. A hard frequency window can change membership when a pole crosses a band edge even if it never crosses the imaginary axis; therefore the boundary-localization theorem below is stated for $\alpha_\perp$. A continuously tracked modal family may be used instead when a family-specific theorem is desired.

IV. MINIMAL REPLACEMENT BLOCKERS AND HYPERGRAPH GEOMETRY

For either stability label $\alpha_\star \in \{\alpha_\perp, \alpha_\Omega\}$ define

$$\mathcal{U}_\star(\theta) = \{S \subseteq \mathcal{V} : \alpha_\star(S, \theta) \geq 0\}, \quad (49)$$

and the inclusion-minimal unstable family

$$\mathcal{H}_\star(\theta) = \min_{\subseteq} \mathcal{U}_\star(\theta), \quad (50)$$

with

$$\kappa_\star(\theta) = \min_{H \in \mathcal{H}_\star(\theta)} |H|. \quad (51)$$

We call each $H \in \mathcal{H}_\star$ a *minimal replacement blocker* (MRB). Unless a mode subcript is shown, $\mathcal{H}$ denotes the complete-spectrum object $\mathcal{H}_\perp$.

Proposition 1 (Antichain and any-order safety): For either label, $\mathcal{H}_\star$ is an antichain. A target portfolio T is stable for every re-equilibrated sequential implementation order, with respect to the chosen label, if and only if T contains no blocker $H \in \mathcal{H}_\star$.

Proof: Minimal elements of any finite family under inclusion form an antichain. If some subset $R \subseteq T$ is unstable, the finite family $\{S \subseteq R : S \in \mathcal{U}_\star\}$ contains an inclusion-minimal element $H \subseteq R$, so $H \in \mathcal{H}_\star$ and $H \subseteq T$. Conversely, if $H \subseteq T$ for some blocker, an implementation order that passes through H includes an unstable stationary configuration. Thus every subset of T is stable if and only if no blocker is contained in T . ■

Theorem 2 (Complete-spectrum boundary localization): Let $\theta(\tau)$ be continuous and suppose g_z remains nonsingular for every portfolio along the path. Then $\mathcal{H}_\perp(\theta(\tau))$ is constant on every interval over which no portfolio has a transverse eigenvalue on the imaginary axis. Every change of $\mathcal{H}_\perp$ is bracketed by such a crossing.

Proof: The Jacobian blocks and therefore $A_\perp(S, \theta(\tau))$ vary continuously while g_z is nonsingular. Eigenvalues vary continuously as a multiset, so the sign of $\alpha_\perp(S, \theta)$ changes only when some transverse eigenvalue satisfies $\text{Re } \lambda = 0$. Since $\mathcal{H}_\perp$ is a deterministic function of the 2^m stability labels, it is constant between such crossings. ■

Remark 1 (Why the hard-band object is different): For $\mathcal{H}_\Omega$, a change can also occur when an already stable or unstable pole crosses the boundary of Ω while retaining the same sign of real part. Thus $\mathcal{H}_\Omega$ is used below as a mechanism-specific diagnostic, while the policy-geometry theorem and any-order planning statements use $\mathcal{H}_\perp$ unless explicitly stated otherwise.

V. FROM THE DAE TO EXACT DYNAMIC MULTIPOINTS

A. Descriptor pencil and device elimination

Linearizing (7)–(8) gives

$$\begin{bmatrix} \Delta \dot{x} \\ 0 \end{bmatrix} = \begin{bmatrix} f_x & f_z \\ g_x & g_z \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta z \end{bmatrix}. \quad (52)$$

The descriptor pencil is

$$P_S(s) = \begin{bmatrix} sI - f_x^{(S)} & -f_z^{(S)} \\ -g_x^{(S)} & -g_z^{(S)} \end{bmatrix}. \quad (53)$$

From the first block row,

$$\Delta x = (sI - f_x)^{-1} f_z \Delta z. \quad (54)$$

Substitute into the algebraic row:

$$T_S(s) \Delta z = 0, \quad T_S(s) = g_z + g_x(sI - f_x)^{-1} f_z. \quad (55)$$

The sign convention is inherited from the frozen code; changing the KCL sign changes corresponding local definitions but not the determinant identities.

The Schur determinant identity gives

$$\det P_S(s) = h_S(s) \det T_S(s), \quad h_S(s) = \det(sI - f_x^{(S)}). \quad (56)$$

Therefore $T_S(s)$ is an exact nonlinear eigenvalue problem (NEP) for network-visible modes away from internal device poles [2].

B. Local SG and GFL dynamic admittances

At a candidate terminal, any linearized device realization can be written

$$\Delta \dot{x}_i = A_i \Delta x_i + B_i \Delta v_i, \quad (57)$$

$$\Delta i_i = C_i \Delta x_i + D_i \Delta v_i. \quad (58)$$

After Laplace transformation,

$$\Delta x_i = (sI - A_i)^{-1} B_i \Delta v_i, \quad (59)$$

so

$$Y_i(s) = D_i + C_i(sI - A_i)^{-1} B_i \quad (60)$$

with $\Delta i_i = Y_i(s) \Delta v_i$. Thus the finite replacement increment is physically

$$\Delta Y_i(s) = Y_i^{\text{GFL}}(s) - Y_i^{\text{SG}}(s), \quad (61)$$

up to the adopted KCL sign. In the implementation, the unambiguous definition is

$$\Delta Y_i(s) = E_i^T [T_{\{i\}}(s) - T_0(s)] E_i \in \mathbb{C}^{2 \times 2}. \quad (62)$$

Proposition 2 (Terminal-map invariance of the collective factor): Consider two internal realizations a and b of every candidate replacement. Suppose, on a protected spectral domain $\mathcal{D}$, (i) their terminal increments are identical, $\Delta Y_i^{(a)}(s) = \Delta Y_i^{(b)}(s)$ for all $s \in \mathcal{D}$, and (ii) neither realization contributes an unrepresented unstable internal pole through the eliminated device factor $h_S(s)$. Then the two realizations generate identical D_S , M_S , Q_S , and collective closure zeros on $\mathcal{D}$.

Proof: The collective construction depends on each replacement only through $\Delta Y_i(s)$ and on the reference grid through $K(s)$. Equality of all terminal increments therefore gives identical D_S , hence identical M_S and Q_S . Assumption (ii) excludes a different unstable zero/pole arising solely from the eliminated factor h_S in (56). ■

Remark 2: This proposition is the route to black-box/vendor use: measured or supplied terminal impedance/admittance scans can feed the same collective calculation without disclosing controller states, provided the terminal model is dynamically complete over the protected band.

C. Base Green kernel and replacement-space operator

Let E select the two rectangular-voltage channels of all m candidate buses. Define

$$K(s) = E^T T_0^{-1}(s) E \quad (63)$$

and, for a portfolio S ,

$$D_S(s) = \text{blkdiag}_{i \in S} \{\Delta Y_i(s)\}. \quad (64)$$

Then

$$M_S(s) = D_S(s) K_{SS}(s). \quad (65)$$

Theorem 3 (Exact network-closure factorization): Whenever $T_0(s)$ is nonsingular and s is not an eliminated device pole,

$$\frac{\det T_S(s)}{\det T_0(s)} = \det(I + M_S(s)). \quad (66)$$

Split $K = K_d + K_o$ into its block diagonal and off-diagonal parts and define

$$Q_S = (I + D_S K_{d,S})^{-1} D_S K_{o,S}. \quad (67)$$

Then

$$\det(I + M_S) = \prod_{i \in S} \det(I + M_{ii}) \det(I + Q_S). \quad (68)$$

Proof: Because each replacement modifies only its own two terminal channels,

$$T_S = T_0 + E_S D_S E_S^T. \quad (69)$$

Apply Sylvester's determinant identity:

$$\det T_S = \det T_0 \det(I + D_S E_S^T T_0^{-1} E_S) \quad (70)$$

$$= \det T_0 \det(I + D_S K_{SS}), \quad (71)$$

which proves (66). Next,

$$I + D_S K_{SS} = I + D_S K_{d,S} + D_S K_{o,S} \quad (72)$$

$$= (I + D_S K_{d,S}) [I + (I + D_S K_{d,S})^{-1} D_S K_{o,S}]. \quad (73)$$

The first factor is block diagonal, so its determinant is the product of the physical local factors. The second factor is $I + Q_S$, proving (68). ■

Corollary 1 (Collective boundary): If every local factor remains nonsingular at s^* ,

$$\det(I + Q_H(s^*)) = 0 \iff -1 \in \text{spec}(Q_H(s^*)). \quad (74)$$

D. Contextual return of each blocker member

Let $H = \{i\} \cup R$ and set $C_H = I + Q_H$. Because the local terms have been dressed out, the diagonal block of Q_H is zero:

$$C_H = \begin{bmatrix} I & Q_{iR} \\ Q_{Ri} & C_R \end{bmatrix}, \quad C_R = I + Q_{RR}. \quad (75)$$

Define

$$\mathcal{R}_{i|R} = Q_{iR} C_R^{-1} Q_{Ri}. \quad (76)$$

Theorem 4 (Contextual-return characterization): If H is an inclusion-minimal blocker at a simple collective root s^* , all local factors are regular, and every proper $C_R(s^*)$ is nonsingular, then

$$1 \in \text{spec}(\mathcal{R}_{i|H \setminus i}(s^*)), \quad \forall i \in H. \quad (77)$$

Moreover, if $C_H q = 0$ and $p^H C_H = 0$, then every local block is nonzero and satisfies

$$\mathcal{R}_{i|R} q_i = q_i, \quad p_i^H \mathcal{R}_{i|R} = p_i^H, \quad (78)$$

so every blocker member participates nontrivially in the critical collective return mode.

Proof: The Schur complement of C_R is

$$I - Q_{iR} C_R^{-1} Q_{Ri} = I - \mathcal{R}_{i|R}, \quad (79)$$

so

$$\det C_H = \det C_R \det(I - \mathcal{R}_{i|R}). \quad (80)$$

At the collective boundary $\det C_H = 0$, while the stated proper-subset regularity gives $\det C_R \neq 0$; hence $\det(I - \mathcal{R}_{i|R}) = 0$ and $+1$ is an eigenvalue. From $C_H q = 0$, the lower block gives $q_R = -C_R^{-1} Q_{Ri} q_i$, and substitution into the upper block gives $(I - \mathcal{R}_{i|R}) q_i = 0$. If $q_i = 0$, then $C_R q_R = 0$; nonsingularity of C_R forces $q_R = 0$, contradicting $q \neq 0$. Thus $q_i \neq 0$ for every $i \in H$. The left-vector result follows analogously from $p^H C_H = 0$. ■

VI. FIVE-BUS WORKED EXAMPLE: EVERY ALGEBRAIC STEP

This section is intentionally pedagogical. It is not the benchmark evidence; it exposes the mechanism with numbers that can be recomputed by hand.

A. Network and Kron reduction

Consider buses 1–4 connected to an internal collector bus 5 with unit susceptance, and bus 5 connected to a strong reference grid by another unit susceptance. Under unit voltages and small angle differences,

$$\Delta P_{ij} \approx b_{ij}(\Delta \delta_i - \Delta \delta_j). \quad (81)$$

The grounded Laplacian is

$$L_5 = \begin{bmatrix} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 \\ -1 & -1 & -1 & -1 & 5 \end{bmatrix}. \quad (82)$$

Setting $\Delta P_5 = 0$ gives

$$\delta_5 = \frac{1}{5}(\delta_1 + \delta_2 + \delta_3 + \delta_4). \quad (83)$$

Hence the exact Kron-reduced stiffness seen at buses 1–4 is

$$L_{\text{red}} = I_4 - \frac{1}{5} \mathbf{1} \mathbf{1}^T. \quad (84)$$

The eigenvalues are $\{1, 1, 1, 0.2\}$; the soft direction is the coherent vector $\mathbf{1}$.

By Sherman–Morrison,

$$K = L_{\text{red}}^{-1} = I_4 + \mathbf{1} \mathbf{1}^T = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{bmatrix}. \quad (85)$$

B. Scalar modal replacement increment

At one selected frequency/direction, represent the SG-to-GFL replacement by the scalar normalized increment

$$\Delta y = -0.2. \quad (86)$$

This value is chosen only to make the closure transparent. For a k -member portfolio, $D = -0.2 I_k$. The physical local factor is

$$I + D K_d = I - 0.4 I = 0.6 I, \quad (87)$$

so no individual replacement is close to local singularity.

For the off-diagonal complete-graph adjacency A_k ,

$$Q_k = (0.6I)^{-1}(-0.2A_k) = -\frac{1}{3}A_k. \quad (88)$$

Because $\text{spec}(A_k) = \{k-1, -1, \dots, -1\}$,

$$\text{spec}(Q_k) = \left\{ -\frac{k-1}{3}, \frac{1}{3}, \dots, \frac{1}{3} \right\}. \quad (89)$$

Thus the collective critical eigenvalue is

$$\lambda_c(I + Q_k) = 1 - \frac{k-1}{3}. \quad (90)$$

Therefore

$$k = 1 : 1, \quad (91)$$

$$k = 2 : \frac{2}{3}, \quad (92)$$

$$k = 3 : \frac{1}{3}, \quad (93)$$

$$k = 4 : 0. \quad (94)$$

The complete four-member portfolio is the first one that closes.

Direct determinants give

$$\det(I + M_k) = \{0.6, 0.32, 0.128, 0\} \quad (95)$$

for $k = 1, 2, 3, 4$, respectively.

C. Contextual return computed by hand

For member 1 and $R = \{2, 3, 4\}$,

$$Q_{1R} = \begin{bmatrix} -1/3 & -1/3 & -1/3 \end{bmatrix} \quad (96)$$

and

$$C_R = \begin{bmatrix} 1 & -1/3 & -1/3 \\ -1/3 & 1 & -1/3 \\ -1/3 & -1/3 & 1 \end{bmatrix}. \quad (97)$$

Its inverse is

$$C_R^{-1} = \begin{bmatrix} 3/2 & 3/4 & 3/4 \\ 3/4 & 3/2 & 3/4 \\ 3/4 & 3/4 & 3/2 \end{bmatrix}. \quad (98)$$

Then

$$C_R^{-1}Q_{R1} = \begin{bmatrix} -1 & -1 & -1 \end{bmatrix}^T \quad (99)$$

and finally

$$\mathcal{R}_{1|234} = Q_{1R}C_R^{-1}Q_{R1} = 1. \quad (100)$$

By symmetry, every member sees unity contextual return at the four-way closure.

VII. FROZEN IEEE-39 EVIDENCE

A. Evidence hierarchy and blind/frozen protocol

The numerical hierarchy is deliberately asymmetric: the complete transverse DAE supplies the portfolio labels; the reduced multiport is asked to predict/explain a selected collective mode; nonlinear phasor time-domain simulation (TDS) checks growth/decay and frequency; independent tools check those blocks that can be reproduced; portfolio-level EMT is not claimed.

For the V4 campaign, the reduced closure construction and its 16 portfolio verdicts were frozen before comparison with

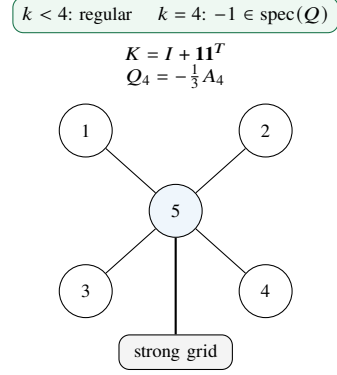


Fig. 1. Five-bus mechanism example. The network itself generates dense transfer coupling after Kron reduction; four locally regular replacements are required before the collective return closes.

TABLE III
EVIDENCE HIERARCHY AND CURRENT SCOPE.

Layer	Role/status
Full transverse DAE	Complete-spectrum ground truth for frozen portfolios
Reduced multiport	Exact model reduction for selected network-visible roots
Phasor nonlinear TDS	32/32 declared small-disturbance sign checks within scope
Independent implementation	Network/equilibrium, synchronous dynamics, and branch sensitivities partially reproduced
EMT	Portfolio-level validation not established

the full-order answer table. This prevents selecting the four-unit witness after seeing the DAE labels. Within the declared phasor-model scope, 32 preregistered small-disturbance TDS verdicts agreed with the eigenvalue signs; for P4 the fitted envelope rate was approximately $+0.129 \text{ s}^{-1}$ versus the linear prediction $+0.127 \text{ s}^{-1}$. Network/equilibrium and synchronous-dynamic blocks have additional cross-tool checks; an independently equivalent realization of the custom GFL is not fully established.

B. The canonical four-unit blocker

At the frozen point $P4 = (g, k, t, h) = (0.03625, 1.425, 1.5, 1)$, the transverse spectral abscissae of the 16 core portfolios are listed in Table IV. All 15 proper subsets are stable, while the complete set is unstable.

The unstable mode is at

$$f^* = 0.6222797 \text{ Hz}, \quad \alpha = +0.1270065 \text{ s}^{-1}. \quad (101)$$

This defines the finite blocker

$$H_4 = \{30, 33, 35, 37\}, \quad \kappa = 4. \quad (102)$$

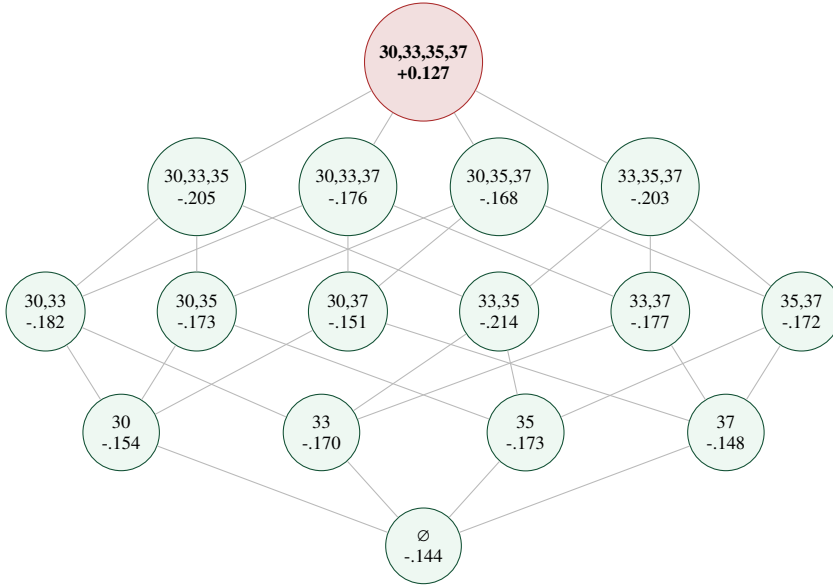
C. Local regularity versus collective collapse

At the clean controller boundary, the minimum physical local singular value is

$$\min_i \sigma_{\min}(I + M_{ii}) = 0.2973, \quad (103)$$

while the collective factor reaches

$$\sigma_{\min}(I + Q_{H_4}) = 1.26 \times 10^{-8}. \quad (104)$$



15/15 proper subsets stable
complete portfolio unstable

Fig. 2. Frozen IEEE-39 core lattice at P4. Numbers are transverse spectral abscissae in s^{-1} . Every proper subset is stable; the complete set is the sole core blocker.

TABLE IV
FROZEN P4 CORE LATTICE ($\alpha_{\perp}$ IN s^{-1}).

S	$\alpha_{\perp}$	S	$\alpha_{\perp}$
$\emptyset$	-0.1441	30+33	-0.1821
30	-0.1542	30+35	-0.1725
33	-0.1696	30+37	-0.1511
35	-0.1734	33+35	-0.2140
37	-0.1480	33+37	-0.1768
30+33+35	-0.2047	35+37	-0.1721
30+33+37	-0.1755	30+35+37	-0.1679
33+35+37	-0.2030	30+33+35+37	+0.1270

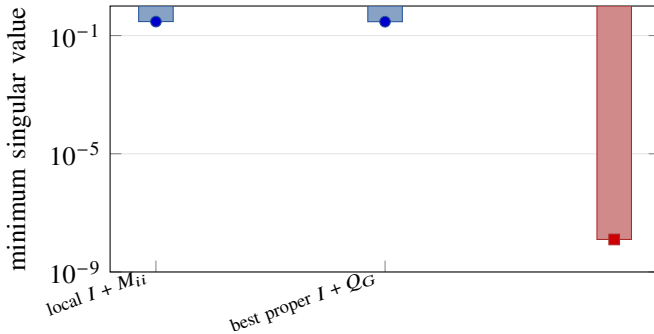


Fig. 3. Physical local factors and the best proper-subset collective factor remain regular while the complete collective factor collapses.

The minimum proper-subset collective factor at the same boundary frequency is 0.2945. Thus the full collective object collapses while every physical local factor and every proper collective subset remain separated from singularity.

At the audited boundary,

$$\min_{\lambda \in \text{spec}(Q_H)} |\lambda + 1| = 3.51 \times 10^{-8}, \quad (105)$$

and the contextual returns for buses 30, 33, 35, 37 approach +1 within approximately 1.1×10^{-7} to 2.3×10^{-7} , with Schur

residuals below 4×10^{-16} .

D. Mode-scoped nine-candidate transfer

For $\mathcal{V}_9 = \{30, \dots, 38\}$ under the same frozen policy, the targeted 0.3–1.5 Hz construction classifies 511/512 portfolios correctly and recovers the mode-scoped blocker antichain $\mathcal{H}_\Omega$. This result is deliberately distinct from the complete-spectrum object $\mathcal{H}_\perp$: a separate family of aperiodic right-half-plane modes produces 110 same-policy full-spectrum false-safe cases for the EM-band predictor. Therefore the collective EM construction is not a complete-spectrum safety certificate.

VIII. COMPATIBILITY IS A GEOMETRY, NOT A PENETRATION NUMBER

The complete-spectrum discrete object $\mathcal{H}_\perp(\theta)$ reorganizes strongly under controller policy. Along one frozen one-dimensional path, the minimum blocker order follows

$$4 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow \emptyset. \quad (106)$$

The corresponding transition values of the GFL Q/V gain are approximately 0.0102, 0.1034, 0.2491, 0.2931, and 0.2973.

Across three two-dimensional policy maps of $\mathcal{H}_\perp$, 30, 36, and 16 distinct blocker hypergraphs were observed. Up to 15 distinct hypergraphs can share the same value of κ ; therefore κ is a useful summary but not the complete planning object.

A particularly important model-scope experiment is that the same P4 four-unit physical portfolio becomes stable when the documented turbine-governor dynamics are added: the transverse abscissa moves from +0.1270 to approximately $-0.0745 s^{-1}$. This removes the specific P4 witness; it does not remove policy dependence in the governed model, where other blockers of orders 2–4 persist. The lesson is that a blocker is conditional on the full dynamic support model, not an intrinsic property of a set of buses.

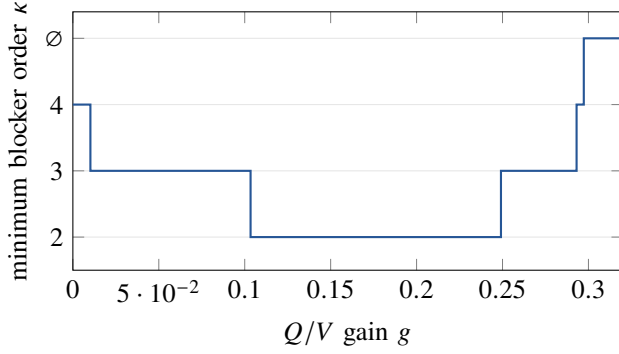


Fig. 4. One-dimensional slice of the compatibility geometry. The same network and replacement locations change their minimum incompatible coalition size as one converter-control coordinate is varied.

Conventional lower-order screens also lose information. At P4, first-order modal sensitivity, additive extrapolation from singleton studies, and pairwise extrapolation from singletons/pairs all predict an empty blocker set. Over 52 policy points they recover the exact blocker hypergraph in 22, 30, and 38 cases, respectively. SCR/gSCR are policy-blind in the constructed P4 versus P_∞ comparison.

IX. A SECONDARY MATHEMATICAL RESULT: FAILURE ORDER IS NOT INTERACTION ORDER

The exact determinant admits Boolean principal-minor coefficients of all orders. However, an algebraic full-order term need not represent an irreducible physical $|S|$ -body interaction. Define

$$F_s(S) = \det(I + Q_{SS}(s)), \quad F_s(\emptyset) = 1. \quad (107)$$

The finite-amplitude partition cumulant is

$$\chi_S(s) = \sum_{\pi \in \Pi(S)} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} F_s(B). \quad (108)$$

Moment-cumulant inversion gives

$$F_s(S) = \sum_{\pi \in \Pi(S)} \prod_{B \in \pi} \chi_B(s). \quad (109)$$

This separates three distinct orders:

$$d_{\text{alg}} = \max\{|S| : \mu_S \neq 0\}, \quad (110)$$

$$d_{\text{conn}} = \max\{|S| : \chi_S \neq 0\}, \quad (111)$$

$$\kappa = \min_{H \in \mathcal{H}} |H|. \quad (112)$$

They are logically independent. In the flagship IEEE-39 boundary, $\kappa = 4$ and full algebraic order is necessary to close the determinant, yet the preregistered connected-share metric is only $\nu_H = 0.069$ and removing the full four-member cumulant moves the root by only about 0.024 s^{-1} . Therefore the flagship is predominantly *composite*, not an irreducible four-body interaction. This negative result prevents an incorrect causal interpretation of the Hasse lattice.

X. EXACT BOUNDARY MOTION AND EXISTING REPAIR RESULTS

Proposition 3 (Simple-root boundary sensitivity): Let $F(s, a) = 0$ describe a simple blocker boundary. Then

$$\frac{\partial s^*}{\partial a} = -\frac{\partial_a F(s^*, a)}{\partial_s F(s^*, a)}. \quad (113)$$

Equivalently, if -1 is a simple eigenvalue of M_S with right/left eigenvectors q, p ,

$$\frac{\partial s^*}{\partial a} = -\frac{p^H (\partial_a M_S) q}{p^H (\partial_s M_S) q}. \quad (114)$$

Proof: Differentiate $F(s^*(a), a) = 0$: $F_s ds^*/da + F_a = 0$. For a simple zero, $F_s \neq 0$, giving (113). For the eigenvalue form, let $\lambda(s, a)$ be the simple eigenvalue of M_S satisfying $\lambda = -1$. Eigenvalue perturbation gives $d\lambda = p^H (dM_S) q / (p^H q)$; impose $d\lambda = 0$ along the boundary and solve for ds/da . ■

In the frozen campaign, port normals agree with direct finite differences to angular error below 4.3×10^{-4} degrees. A port-derivative Newton continuation moves the four-unit P4 portfolio to the Q/V stability boundary $g^* = 0.20768$ with approximately 2×10^{-10} final error in five iterations. A damped synchronous condenser restores small-signal stability above approximately 2.48% of the retired rating, about 106 MVA in that study. These are existing validated repair results.

XI. BLOCKER-CONDITIONED CRITICAL GREEN RESIDUE AND REPAIR GRAPH

This section derives the spatial design extension. It corrects an important subtlety: at a critical root s^* , $T(s^*)$ is singular, so the ordinary inverse $T^{-1}(s^*)$ does not exist. The correct critical Green object is the *residue* of the resolvent. The residue-sensitivity equivalence has prior art in impedance participation analysis [5]; the contribution pursued here is its blocker-conditioned use together with exact collective closure, total re-equilibrated topology derivatives, and hypergraph-aware repair.

A. Self-energy network operator and generalized branch stamps

Partition retained network variables from hidden controller/auxiliary states. Exact Schur condensation gives

$$S_r(s) = sI - A_{rr} - A_{rc}(sI - A_{cc})^{-1}A_{cr}. \quad (115)$$

When retained coordinates admit a second-order interpretation,

$$\begin{aligned} T(s) &= s^2 M + s \Sigma(s) + L, \\ \Sigma(s) &= D_0 + C(sI - A_{cc})^{-1}B. \end{aligned} \quad (116)$$

For a network with lines, transformers, shunts, loads, and condensed devices, use generalized terminal stamps

$$T(s) = \sum_{e \in \mathcal{E}} C_e^H Y_e(s) C_e + Y_{\text{sh}}(s) + \Sigma_{\text{dev}}(s). \quad (117)$$

Here C_e maps bus-voltage coordinates to the two branch-terminal coordinates and can include off-nominal transformer ratios and phase shifts. For a symmetric ordinary line, C_e reduces to the usual incidence difference $(e_i - e_j)^T \otimes I_2$.

B. Critical Green residue

Let s_H^* be a simple network-visible root of portfolio H :

$$T_H(s_H^*)r_H = 0, \quad \ell_H^H T_H(s_H^*) = 0, \quad (118)$$

with

$$d_H^* = \ell_H^H T_{H,s}(s_H^*)r_H \neq 0. \quad (119)$$

Near the simple pole, the resolvent has the Laurent expansion

$$T_H^{-1}(s) = \frac{\mathcal{G}_H^*}{s - s_H^*} + G_{H,\text{reg}}(s), \quad \mathcal{G}_H^* = \frac{r_H \ell_H^H}{d_H^*}. \quad (120)$$

Thus $\mathcal{G}_H^*$ is a rank-one critical Green residue. For any differentiable perturbation direction T_a evaluated at a fixed equilibrium,

$$\frac{\partial s_H^*}{\partial a} = -\frac{\ell_H^H T_a r_H}{d_H^*} = -\text{tr}(\mathcal{G}_H^* T_a). \quad (121)$$

A scale-invariant conditioning indicator is

$$\kappa_{\text{NEP}}(s_H^*) = \frac{\|\ell_H\|_2 \|r_H\|_2}{|\ell_H^H T_{H,s} r_H|}. \quad (122)$$

Large κ_{NEP} warns that repair rankings derived from first-order pole motion may be fragile to modeling or numerical errors.

C. Total topology derivative: structural, flow, and controller channels

A finite branch change also moves the AC equilibrium. Let ξ collect the equilibrium variables and let

$$\mathcal{F}(\xi, \rho) = 0 \quad (123)$$

be the equilibrium equations for a scalar branch parameter ρ . If $\mathcal{F}_\xi$ is nonsingular,

$$\xi_\rho = -\mathcal{F}_\xi^{-1} \mathcal{F}_\rho. \quad (124)$$

The total derivative of the dynamic operator is therefore

$$\widehat{T}_\rho = \partial_\rho T + D_\xi T[\xi_\rho] = \partial_\rho T - D_\xi T[\mathcal{F}_\xi^{-1} \mathcal{F}_\rho]. \quad (125)$$

For a branch action it is useful to decompose

$$\widehat{T}_{\rho_e} = \underbrace{T_{\rho_e}^{\text{dir}}}_{\text{structural}} + \underbrace{T_{\rho_e}^{\text{flow}}}_{\text{power-flow mediated}} + \underbrace{T_{\rho_e}^{\text{ctrl}}}_{\text{controller mediated}}, \quad (126)$$

where the latter two terms arise through the re-equilibrated state ξ_ρ . The re-equilibrated pole sensitivity is

$$\frac{ds_H^*}{d\rho_e} = -\text{tr}(\mathcal{G}_H^* \widehat{T}_{\rho_e}). \quad (127)$$

Equation (127), rather than the direct branch stamp alone, is the quantity that should be compared against finite branch doubling/outage experiments.

D. Parameterization-invariant line leverage

Raw derivatives depend on the units used for ρ_e . For a positive branch-strength parameter define $\eta_e = \log \rho_e$. Then

$$\frac{\partial s_H^*}{\partial \eta_e} = \rho_e \frac{\partial s_H^*}{\partial \rho_e}. \quad (128)$$

The dimensionally consistent stabilizing score for a fractional reinforcement is

$$\widehat{S}_{e|H}^* = -\text{Re} \left\{ \frac{\partial s_H^*}{\partial \eta_e} \right\} = \text{Re} \left\{ \rho_e \text{tr}(\mathcal{G}_H^* \widehat{T}_{\rho_e}) \right\}. \quad (129)$$

Positive $\widehat{S}_{e|H}^*$ means that a small percentage increase of the selected branch-strength parameter moves the blocker root left. A cost-normalized version divides the predicted finite displacement by the engineering cost of the admissible action.

For an ordinary symmetric line and a frozen equilibrium, $\partial_{\rho_e} T = C_e^H Y_{e,\rho} C_e$, and (121) reduces to the familiar left/right edge-field product. The full score (129) retains the additional power-flow and controller-mediated channels.

E. Bus/control leverage and dynamic hubs

For a bus-local controller parameter θ_i , define its total operator derivative $\widehat{T}_{\theta_i}$ analogously to (125). Then

$$\frac{ds_H^*}{d\theta_i} = -\text{tr}(\mathcal{G}_H^* \widehat{T}_{\theta_i}). \quad (130)$$

A practical node score should be normalized by an admissible parameter change (or cost), rather than by arbitrary parameter units. For visualization only, incident absolute edge leverage can be aggregated as

$$H_i^* = \sum_{e \sim i} |\widehat{S}_{e|H}^*|. \quad (131)$$

This “dynamic hub” is not claimed as a universal graph centrality.

F. Dynamic weak zones via exterior self-energy

Partition retained network variables into a zone Z and exterior $\bar{Z}$:

$$T = \begin{bmatrix} T_{ZZ} & T_{Z\bar{Z}} \\ T_{\bar{Z}Z} & T_{\bar{Z}\bar{Z}} \end{bmatrix}. \quad (132)$$

When $T_{\bar{Z}\bar{Z}}$ is nonsingular, eliminating the exterior gives

$$T_Z^{\text{eff}} = T_{ZZ} - \Sigma_Z^{\text{ext}}, \quad \Sigma_Z^{\text{ext}} = T_{Z\bar{Z}} T_{\bar{Z}\bar{Z}}^{-1} T_{\bar{Z}Z}. \quad (133)$$

Introduce the diagnostic interpolation $T_Z(\eta) = T_{ZZ} - \eta \Sigma_Z^{\text{ext}}$. At a simple zone root,

$$\frac{ds^*}{d\eta} = \frac{\ell_Z^H \Sigma_Z^{\text{ext}}(s^*) r_Z}{\ell_Z^H \partial_s T_Z^{\text{eff}}(s^*) r_Z}. \quad (134)$$

Its real part diagnoses whether the rest of the grid locally supplies stabilizing or destabilizing dynamic feedback to the selected zone; η is a diagnostic interpolation, not by itself a physical switching action.

G. Complete modal balance over graph and nodal terms

At a simple root, premultiplying (117) by ℓ^H and postmultiplying by r gives the exact complex balance

$$\sum_{e \in \mathcal{E}} (C_e \ell)^H Y_e(s^*) (C_e r) + \ell^H [Y_{sh}(s^*) + \Sigma_{dev}(s^*)] r = 0. \quad (135)$$

The second term includes shunts/load Jacobians and device/controller self-energy; omitting it would give an incomplete balance. Equation (135) is a model-based modal accounting identity, not a substitute for measurement-based dissipating-energy-flow source localization [12].

H. Finite low-rank update: exact only at a frozen equilibrium

For s such that $T(s)$ is nonsingular, a frozen-operating-point low-rank branch modification

$$T^+(s) = T(s) + C_e^H \Delta Y_e(s) C_e \quad (136)$$

has the exact Woodbury update

$$G^+ = G - G C_e^H \left(\Delta Y_e^{-1} + C_e G C_e^H \right)^{-1} C_e G, \quad (137)$$

and determinant ratio

$$\frac{\det T^+(s)}{\det T(s)} = \det \left[I + \Delta Y_e(s) C_e G(s) C_e^H \right]. \quad (138)$$

At $s = s^*$, $G = T^{-1}$ is singular and the residue (120) is the correct critical object. Equations (137)–(138) are exact for a frozen equilibrium; a physical branch doubling/outage must be re-equilibrated and relinearized before its finite final root is certified.

I. Hypergraph-aware minimal repair

Repairing one blocker while creating another is not acceptable. Let $\{H_k\}_{k=1}^q$ be the blockers/near-boundary portfolios selected for protection, with critical roots s_k^* , and let $x_e = \Delta \eta_e$ be relative branch changes. Define

$$J_{ke} = \text{Re} \left\{ \frac{\partial s_k^*}{\partial \eta_e} \right\}. \quad (139)$$

A first-order topology design is

$$\min_x \sum_e c_e |x_e| \quad (140)$$

$$\text{s.t.} \quad \text{Re } s_k^* + \sum_e J_{ke} x_e \leq -\varepsilon, \quad k = 1, \dots, q, \quad (141)$$

$$x \in \mathcal{X}_{\text{adm}}. \quad (142)$$

Controller variables can be appended using (130). The program is a proposal generator, not a global certificate: every selected finite action is subsequently re-equilibrated, relinearized, and the protected portfolio family is re-enumerated to ensure that the repair did not create a smaller blocker elsewhere.

XII. TOPOLOGY EVIDENCE AND THE PROPOSED BLIND GREEN-RESIDUE TEST

A frozen finite-action campaign already establishes that topology can both repair and create incompatibility. At P4, four single branch doublings (campaign indices 0, 1, 13, and 43) independently drive the H_4 abscissa negative while every proper subset is already stable. Conversely, some outages create new smaller blockers, including cases with $\kappa = 1$. Across 810 tested policy/action pairs, the complete-spectrum blocker hypergraph changes in 169 cases. Static changes in Fiedler value, Kirchhoff index, gSCR, and minimum SCR did not consistently predict the effect.

The next experiment should be preregistered before the finite labels are consulted:

- 1) Freeze P4, H_4 , the equilibrium, and the critical root s_H^* .
- 2) Compute r_H , ℓ_H , $\mathcal{G}_H^*$, and κ_{NEP} once.
- 3) For each admissible branch, compute both the direct/frozen score and the total re-equilibrated score (129), preserving the structural/flow/controller decomposition (126).
- 4) Rank branches without using finite doubling/outage labels.
- 5) Reveal the finite labels and report sign accuracy, Spearman correlation with $\Delta \alpha_{\perp}$, recall@4 for the four known P4 repairs, and precision-recall AUC because the positive repair class is sparse.
- 6) Re-evaluate the highest-ranked proposed repairs across the complete protected portfolio family to test whether a repaired blocker is replaced by a new smaller blocker.

The strongest possible outcome is not merely a high rank correlation. It is a mechanistic separation: a static/direct topology score fails, while the total Green-residue score succeeds because power-flow redistribution and controller self-energy materially change the line ranking. A negative result is equally informative and should remain in the record without retuning.

XIII. WHAT THE POSTER CAN CLAIM AND WHAT IT CANNOT

The strongest defensible chain is

$$\begin{aligned} \text{WHO fails?} &\rightarrow \mathcal{H}_{\perp}, \mathcal{H}_{\Omega} \\ \text{WHY?} &\rightarrow Q_H, \mathcal{R}_{i|R} \\ \text{WHERE is leverage?} &\rightarrow \mathcal{G}_H^* \\ \text{WHAT should change?} &\rightarrow ds_H^*/d\rho, ds_H^*/d\theta. \end{aligned} \quad (143)$$

Supported within the frozen model scope are:

- a finite four-unit complete-spectrum blocker exists at P4, and its critical root lies in the targeted EM family;
- its EM boundary is a collective closure while physical local factors remain regular;
- a critical eigenvalue of Q_H approaches -1 , while each contextual return approaches $+1$ and every blocker member has nonzero critical local support;
- the complete-spectrum blocker hypergraph reorganizes with controller policy, including the observed $4 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow \emptyset$ sequence;
- exact port derivatives move the boundary and support successful iterative Q/V retuning;

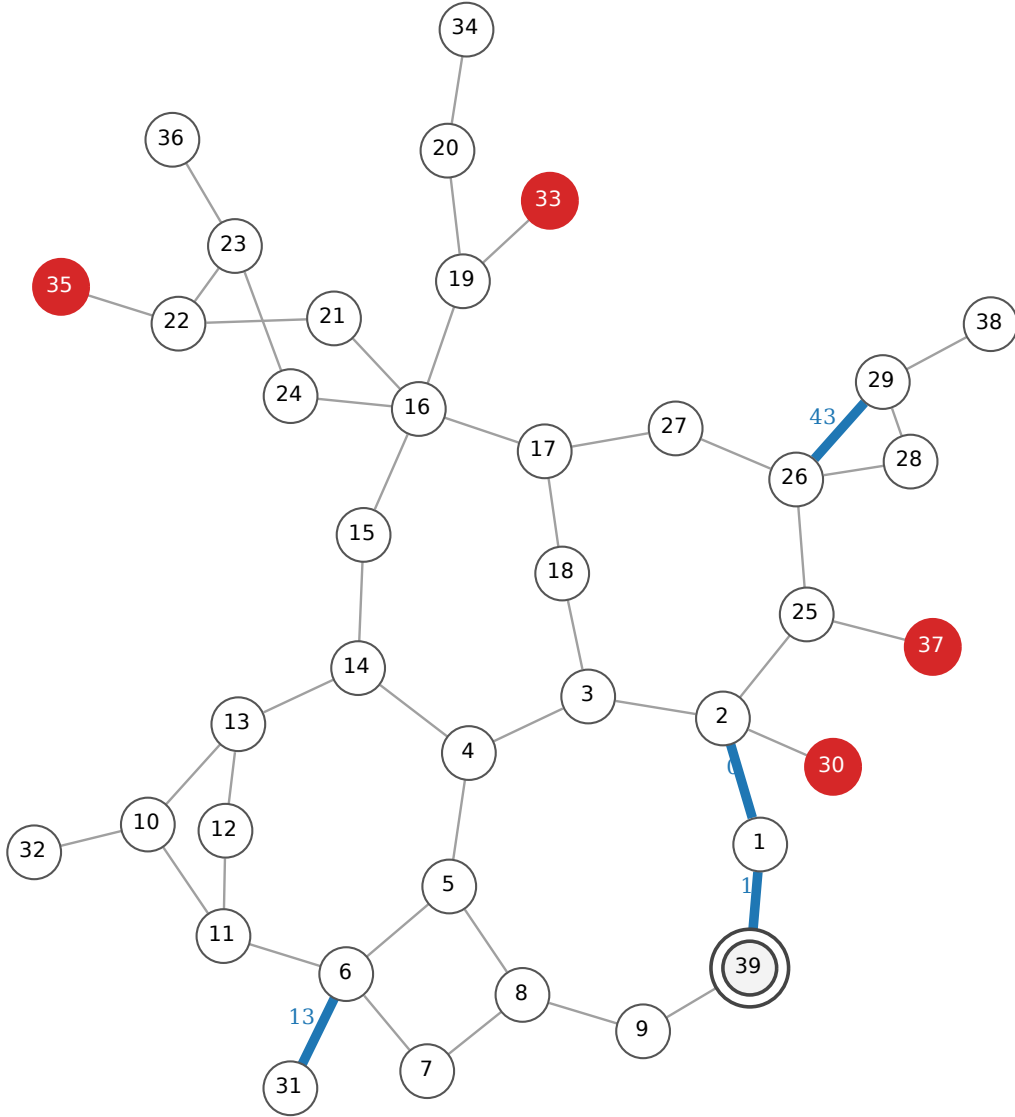


Fig. 5. IEEE-39 topology with the four candidate blocker buses (red) and the four branch reinforcements that independently repaired P4 in the frozen finite-action campaign (blue; campaign branch indices 0, 1, 13, and 43). This is known repair evidence, not a Green-residue prediction. The proposed blind experiment ranks all admissible branches from one critical blocker mode using (129) and only then reveals the finite-action labels.

- finite topology and synchronous-support actions can repair the benchmark blocker;
- the V4 mechanism was frozen/blind before full-order reveal and is corroborated by nonlinear phasor TDS within the declared model scope.

The following are explicitly not claimed:

- complete-spectrum safety from the 0.3–1.5 Hz predictor;
- universal order-four or irreducible four-body physics;
- a simple physical cycle/holonomy as the blocker cause;
- portfolio-level EMT validation or manufacturer-model universality;
- validated Green-residue branch ranking before the preregistered branch experiment is run;
- scalability of exact portfolio enumeration to very large candidate sets without a pre-screening/search layer.

XIV. POSTER AND PRESENTATION EXTRACTION

The full theory should not be printed on the poster. A four-panel extraction is:

- 1) **Discover – WHO fails?** Fig. 2; “15 proper portfolios stable, complete portfolio unstable.” State that this P4 blocker is complete-spectrum and that its critical pole lies at 0.622 Hz.
- 2) **Explain – WHY?** Eq. (68), Fig. 3, and the audited $Q_H \rightarrow -1$, $\mathcal{R}_{i|R} \rightarrow +1$ return closure.
- 3) **Map – HOW does compatibility move?** Fig. 4; “compatibility is a controller-dependent combinatorial geometry,” with $\mathcal{H}_\perp$ clearly distinguished from the EM diagnostic $\mathcal{H}_\Omega$.
- 4) **Repair – WHERE and WHAT?** Existing Q/V/condenser/topology repairs plus Fig. 5. If the blind Green-residue ranking succeeds, overlay the

normalized total score $\hat{S}_{e|H}^*$ and its structural/flow/controller decomposition directly on the grid.

The one equation to emphasize in an expert conversation is

$$\frac{ds_H^*}{d\rho_e} = -\text{tr}\left[\mathcal{G}_H^* \left(\partial_{\rho_e} T - D_\xi T[\mathcal{F}_\xi^{-1} \mathcal{F}_{\rho_e}]\right)\right]. \quad (144)$$

It unifies the blocker-conditioned critical mode, Green residue, topology perturbation, power-flow redistribution, controller self-energy, and design direction. The five-bus derivation remains presentation backup material for explaining the collective-return mechanism without the full IEEE-39 DAE.

XV. CONCLUSION

The theory developed here separates four questions that are often conflated in SG-to-GFL transition studies. The complete-spectrum blocker hypergraph identifies *which* finite portfolios first fail; the exact dynamic multiport and contextual return explain *why* a selected blocker closes collectively; the policy maps show that compatibility is a controller-dependent combinatorial geometry rather than a scalar penetration threshold; and the critical Green residue provides a mathematically well-defined bridge from the singular mode to spatial repair sensitivities. The most important pending experiment is deliberately narrow: rank the frozen IEEE-39 branch actions from one blocker mode using the total re-equilibrated residue score, reveal the known finite repair labels only afterward, and verify that the selected action repairs the protected hypergraph rather than merely one eigenvalue. Success would provide a compact diagnose–localize–repair chain; failure would sharply delimit the value of modal repair fields. In either case, complete-spectrum labels, model scope, and the distinction between classical algebra and new interpretation must remain explicit.

APPENDIX A

PRINCIPAL-MINOR EXPANSION AND WHY FULL ALGEBRAIC ORDER NEED NOT MEAN FULL PHYSICAL INTERACTION

For any square Q ,

$$\det(I + Q) = \sum_U \det Q[U, U], \quad (145)$$

where the empty principal minor is one. Grouping channel-level minors by the set of device blocks they touch gives

$$\det(I + Q_{SS}) = \sum_{T \subseteq S} \mu_T, \quad (146)$$

with $\mu_\emptyset = 1$ and, because $Q_{ii} = 0$, $\mu_{\{i\}} = 0$. This exact algebraic expansion explains why a truncation below $|S|$ can fail even when the irreducibly connected cumulant of order $|S|$ is small: products of lower-order connected structures can require the full support S to assemble.

APPENDIX B

STATIC SYMMETRIC LIMIT OF THE CRITICAL EDGE SCORE

Suppose the retained operator reduces to the real symmetric Laplacian eigenproblem $L(\rho)v = \lambda v$ with $v^T v = 1$, and a

line weight enters as $L = L_0 + w_e b_e b_e^T$. Standard perturbation theory gives

$$\frac{\partial \lambda}{\partial w_e} = v^T b_e b_e^T v = (b_e^T v)^2. \quad (147)$$

The blocker-conditioned formulas above are the dynamic non-Hermitian/NEP generalization: the single symmetric mode is replaced by distinct left/right fields, the normalization by $\ell^H T_s r$, and the direct line stamp by the total re-equilibrated operator derivative. Away from a pole, $C_e T^{-1}(s) C_e^H$ is a matrix-valued dynamic effective-resistance object; in the static grounded/pseudoinverse limit it reduces to $b_e^T L^\dagger b_e$. Thus Fiedler/effective-resistance quantities appear as special static symmetric limits, not as universal dynamic repair scores.

REFERENCES

- [1] N. Hatziaargyriou *et al.*, “Definition and classification of power system stability—revisited & extended,” *IEEE Trans. Power Syst.*, vol. 36, no. 4, pp. 3271–3281, 2021.
- [2] F. Tisseur and K. Meerbergen, “The quadratic eigenvalue problem,” *SIAM Review*, vol. 43, no. 2, pp. 235–286, 2001.
- [3] W. Dong, H. Xin, D. Wu, and L. Huang, “Small signal stability analysis of multi-infeed power electronic systems based on grid strength assessment,” *IEEE Trans. Power Syst.*, vol. 34, no. 2, pp. 1393–1403, 2019.
- [4] A. Gorbunov, J. C.-H. Peng, and P. Vorobev, “Identification of critical clusters in inverter-based microgrids,” *Electr. Power Syst. Res.*, vol. 189, Art. no. 106731, 2020.
- [5] Y. Zhu, Y. Gu, Y. Li, and T. C. Green, “Participation analysis in impedance models: The grey-box approach for power system stability,” *IEEE Trans. Power Syst.*, vol. 37, no. 1, pp. 343–353, Jan. 2022, doi: 10.1109/TPWRS.2021.3088345.
- [6] Y. Zhu, Y. Gu, Y. Li, and T. C. Green, “Impedance-based root-cause analysis: Comparative study of impedance models and calculation of eigenvalue sensitivity,” *IEEE Trans. Power Syst.*, vol. 38, no. 2, pp. 1642–1654, 2023.
- [7] L. Huang, D. Wang, X. Wang, H. Xin, P. Ju, K. H. Johansson, and F. Dörfler, “Gain and phase: Decentralized stability conditions for power electronics-dominated power systems,” *IEEE Trans. Power Syst.*, vol. 39, no. 6, pp. 7240–7256, 2024.
- [8] F. Conte, F. Mancilla-David, F. Silvestro, and S. Grillo, “Small-signal stability manifolds in converter-dominated power systems,” *arXiv:2605.26254*, 2026.
- [9] T. Joswig-Jones, S. Dong, J. Tan, and B. Zhang, “Sensitivity-based system strength assessment: Mapping power flow and network topology perturbations to system eigenvalues,” *arXiv:2607.28764*, 2026.
- [10] A. G. J. MacFarlane, “Return-difference and return-ratio matrices and their use in analysis and design of multivariable feedback control systems,” *Proc. IEE*, vol. 117, no. 10, pp. 2037–2049, 1970.
- [11] A. G. J. MacFarlane and I. Postlethwaite, “The generalized Nyquist stability criterion and multivariable root loci,” *Int. J. Control*, vol. 25, no. 1, pp. 81–127, 1977.
- [12] S. Maslennikov, B. Wang, and E. Litvinov, “Dissipating energy flow method for locating the source of sustained oscillations,” *Int. J. Electr. Power Energy Syst.*, vol. 88, pp. 55–62, 2017.